1 Result in short

A LightGBM model ranks the stocks of the S&P 500 every Friday by their expected return over the next week, relative to their sector. Each year from 2013 to 2022 is predicted by a model fitted on earlier years only, but the settings of the model and of the portfolio were chosen while looking at this period, so 2013-2022 is a development period, not a clean out-of-sample one. The period 2023-2026 was not used to build the model.

The weekly rank correlation between predictions and realised sector-relative returns (the IC, section 3) has a mean of 0.0161 over 2013-2022 ($t = 4.06$) and 0.0135 over 2023-2026 ($t = 1.89$). Most of it comes from effects every quantitative fund already knows, above all short-term reversal: a simple reversal rule (0.0190) and a ridge regression (0.0198) even have a higher IC over 2013-2022. LightGBM keeps the most once sector, beta, size, 5- and 21-day reversal, momentum and volatility are removed, with pure IC of 0.0081, $t = 3.23$ over 2013-2022 and 0.0050, $t = 1.14$ over 2023-2026 (37% of it).

Beyond known effects, nothing significant is left since 2020. Once the one-day reversal and the other short-term features are removed as well (section 3), the pure IC is about 0.008 over 2013-2019, about 0 over 2020-2022 and about 0.004 ($t \approx 1$) over 2023-2026. What remains comes almost entirely from 2013-2019, the years in which the data miss the most stocks (only 71% of the index members have a price in 2013, against 92% in 2022). Looks like survivorship bias unfortunately.

A beta-hedged, dollar-neutral portfolio built on it has a Sharpe ratio of 0.42 before costs over 2013-2022, which is not significantly different from zero ($t = 1.39$), and breaks even at a transaction cost of about 6 basis points per trade.

2 Data and notation

Let $\mathcal{T}$ be the ordered set of sessions from 3 January 2005 on which SPY (the ETF that tracks the S&P 500) traded. For a session $t \in \mathcal{T}$ and an integer k, $t + k$ denotes the k-th session after t. For a stock i and a session t:

- $O_{i,t}$, $C_{i,t}$, $V_{i,t}$ are the opening price, the closing price and the traded volume, adjusted for splits and dividends.
- $M_{i,t} \in \{0, 1\}$ is 1 if i is a member of the S&P 500 on t, from the historical list of members maintained in `github.com/fja05680/sp500`.
- C^m_t, O^m_t are the close and open of SPY, and $C^{(k)}_t$ the close of the SPDR sector ETF k.

we use 2 kinds of daily returns, that is the log return $\ell_{i,t} = \ln(C_{i,t}/C_{i,t-1})$ (and ℓ^m_t for SPY), for the features, and the simple return $r_{i,t} = C_{i,t}/C_{i,t-1} - 1$ (and r^m_t, $r^{(k)}_t$ for SPY and the sector ETFs), for the sectors and the market hedge.

The data contain every stock that was a member of the index at some date between 2005 and October 2026, whether or not it is a member on a given day. Yahoo Finance has no prices for most companies that left the index so only 71% of the members have a price in 2013, 81% in 2018 and 92% in 2022, so survivorship bias is a problem for us.

A *week-end session* is the last session of a calendar week, $\mathcal{W} \subset \mathcal{T}$ is the set of week-end sessions, and for $t \in \mathcal{W}$, the *universe* is defined by

$$\mathcal{U}_t = \{\, i : M_{i,t} = 1,\ C_{i,t} \text{ observed},\ C_{i,t-252} \text{ observed} \,\}.$$

Sectors. No historical industry classification is freely available, so the sector of a stock is the

sector ETF whose daily returns had the highest correlation with the stock's over the last 252 sessions. Number the sessions of $\mathcal{T}$ from 0 and let $a_0 < a_1 < \dots$ be the sessions with numbers $252 + 21\mathbb{N}$. On each a_j,

$$s_i(a_j) := \arg\max_k \ \operatorname{corr}\big(r_{i,a_j-251..a_j},\, r^{(k)}_{a_j-251..a_j}\big),$$

over the ETFs k with at least 126 returns in common with the stock (so XLRE and XLC can be chosen only about six months after their start). The label is carried forward: $s_i(t) = s_i(a_j)$ for the last $a_j \le t$.

3 Features

The 19 raw features of stock i at session t are listed below.

name	definition	meaning
`rev_1`	$-\ell_{i,t}$	1-day reversal
`rev_5`	$-(\ln C_{i,t} - \ln C_{i,t-5})$	1-week reversal
`rev_21`	$-(\ln C_{i,t} - \ln C_{i,t-21})$	1-month reversal
`mom_6_1`	$\ln C_{i,t-21} - \ln C_{i,t-126}$	6-month momentum, last month skipped
`mom_12_1`	$\ln C_{i,t-21} - \ln C_{i,t-252}$	12-month momentum, last month skipped
`rev_5_sector`	rev_5 minus its sector average A_t (below)	sector-relative reversal
`mom_12_1_sector`	mom_12_1 minus its sector average A_t	sector-relative momentum
`vol_21, vol_63`	$\mathrm{sd}_{21}(\ell_i)$, $\mathrm{sd}_{63}(\ell_i)$	volatility
`beta_252`	$\widehat{\beta}_{i,t} = \widehat{\mathrm{Cov}}_{252}(\ell_i, \ell^m)/\widehat{\mathrm{Var}}_{252}(\ell^m)$	market beta
`idio_vol_63`	$\big(\widehat{\mathrm{Var}}_{63}(\ell_i) - \widehat{\beta}^2_{i,t}\,\widehat{\mathrm{Var}}_{63}(\ell^m)\big)^{1/2}_+$	stock-specific volatility
`max_21`	$\max_{21}(\ell_i)$	largest daily gain
`skew_63`	$\mathrm{skew}_{63}(\ell_i)$	asymmetry of returns
`dist_52w_high`	$\ln C_{i,t} - \max_{252}(\ln C_i)$	distance to the 1-year high
`dollar_vol_21`	$\ln \mathrm{mean}_{21}(C_i V_i)$	size and liquidity
`vol_shock`	$\ln\big(V_{i,t}/\,\mathrm{mean}_{21}(V_i)\big)$	unusual volume
`gap`	$\ln(O_{i,t}/C_{i,t-1})$	overnight move
`intraday`	$\ln(C_{i,t}/O_{i,t})$	move during the day
`rsi2`	$100 - 100/(1 + U_{i,t}/D_{i,t})$, with $U_{i,t} = \frac{1}{2}U_{i,t-1} + \frac{1}{2}(\ell_{i,t})_+$ and $D_{i,t} = \frac{1}{2}D_{i,t-1} + \frac{1}{2}(-\ell_{i,t})_+$	2-day relative strength

Sector average. For a feature x, $A_t(x)_i$ is the mean of $x_{j,t}$ over all stocks j of the data with $s_j(t) = s_i(t)$ and $x_{j,t}$ observed, members of the index on t or not. On an average day of 2013-2022 the average runs over about 600 stocks, of which about 405 are members. Because the data contain every stock that was a member up to October 2026, the averages also include 58 stocks that joined the index only after 2022 (about 8% of the stocks averaged), hich is a small use of information from after the development period. Restricting the averages to the members of the day would change the cross-sectional ranks of `rev_5_sector` and `mom_12_1_sector` only a little (mean rank correlation 0.98 with the version used). The frozen live code computes the features this way, so the definition is kept and stated here.

Cross-sectional rank

For a variable x defined on $\mathcal{U}_t$, let n_t be the number of stocks of $\mathcal{U}_t$ where x is observed and $\mathrm{rank}_t(x_i) \in [1, n_t]$ the rank of x_i among these n_t values (tied values receive the average of the ranks they occupy, so a rank can be a half-integer). The ranked variable is

$$R_t(x)_i = \frac{\mathrm{rank}_t(x_i)}{n_t} - \frac{1}{2} \in \left(-\tfrac{1}{2}, \tfrac{1}{2}\right].$$

For the features, a missing value is set to $R_t(x)_i = 0$, the middle of the ranking. For the target, a missing value stays missing and the stock is left out of that week.

The model input for stock i at week t is the vector $X_{i,t} \in \mathbb{R}^{19}$ of the ranked features.

For $t \in \mathcal{W}$ and $i \in \mathcal{U}_t$, the forward return is measured from the next open to the open five sessions later, which is from Monday's open to the following Monday's open in a normal week:

$$y_{i,t} = \ln O_{i,t+6} - \ln O_{i,t+1}.$$

It is made relative to the sector and then ranked:

$$\tilde{y}_{i,t} = y_{i,t} - \frac{1}{|S_{i,t}|} \sum_{j \in S_{i,t}} y_{j,t}, \qquad S_{i,t} = \{\, j \in \mathcal{U}_t : s_j(t) = s_i(t),\ y_{j,t} \text{ observed} \,\},$$

$$Z_{i,t} = R_t(\tilde{y})_i.$$

$Z_{i,t}$ is our target. It is not known at t: it depends on prices from $t+1$ to $t+6$. Unlike the feature averages, the sector average here is taken over the universe $\mathcal{U}_t$ only.

Ridge regression:

$$\widehat{\theta} = \arg\min_{\theta \in \mathbb{R}^{19},\, c \in \mathbb{R}} \sum_{(i,t)} \big(Z_{i,t} - c - X_{i,t}^{\top}\theta\big)^2 + \alpha \|\theta\|^2, \qquad \alpha = 10,$$

with prediction $\widehat{c} + X^{\top}\widehat{\theta}$. A ranked feature has variance close to $1/12$, so with n training rows the penalty $\alpha = 10$ is negligible next to $n\,\mathrm{Var}(X_j) \approx n/12$, which is above 8,000 for every fit: in practice this ridge is ordinary least squares.

LightGBM, gradient-boosted regression trees from the LightGBM library . Starting from $F_0 = \bar{Z}$,

$$F_m(x) = F_{m-1}(x) + \eta\, h_m(x), \quad m = 1, \ldots, 300,$$

where h_m is a tree fitted to the residuals $Z_{i,t} - F_{m-1}(X_{i,t})$ of the current model (squared loss), and $\eta = 0.03$. Each tree has a depth of at most 5 and at most 31 leaves, every leaf holds at least 2,000 stock-weeks, and each tree is fitted on a random 70% of the rows and of the features (random seed 0). These settings were fixed during development, on 2013-2022.

Training and prediction

For each year Y, let τ_Y be the third-to-last week-end session before 1 January of Y. The model used in year Y is fitted on all pairs (i,t) with $t \in \mathcal{W}$, $2008 \le t \le \tau_Y$, $i \in \mathcal{U}_t$ and $Z_{i,t}$ observed, and then predicts every week of Y. The two weeks left out ensure that no training target overlaps year Y.

Information coefficient

For a week t, with predictions $p_{\cdot,t}$ and targets $Z_{\cdot,t}$ on the stocks of $\mathcal{U}_t$ where both are defined,

$$\mathrm{IC}_t = \mathrm{corr}\big(R_t(p_{\cdot,t}),\, Z_{\cdot,t}\big).$$

Over a set of weeks W with $n = |W|$, the mean IC is $\overline{\mathrm{IC}} = \frac{1}{n}\sum_{t \in W} \mathrm{IC}_t$.

IC_t is a random variable: it depends on the returns of week t. The statement "the model has information" is the statement $\mu = \mathbb{E}[\mathrm{IC}_t] > 0$, tested with the statistic below.

Let $e_t = \mathrm{IC}_t - \overline{\mathrm{IC}}$ and $\widehat{\gamma}_k = \frac{1}{n}\sum_t e_t e_{t-k}$ (sum over the pairs of weeks of W that are k weeks apart). With L lags,

$$\widehat{V} = \widehat{\gamma}_0 + 2\sum_{k=1}^{L}\Big(1 - \frac{k}{L+1}\Big)\widehat{\gamma}_k, \qquad t_{NW} = \frac{\overline{\mathrm{IC}}}{\sqrt{\widehat{V}/n}}.$$

	2013-2022 ($n = 520$)		2013-2019		2020-2022	
model	$\overline{\mathrm{IC}}$	t_{NW}	$\overline{\mathrm{IC}}$	t_{NW}	$\overline{\mathrm{IC}}$	t_{NW}
reversal rule*	0.0190	3.13	0.0155	2.49	0.0272	1.95
ridge	0.0198	3.69	0.0130	2.29	0.0358	3.11
LightGBM	0.0161	4.06	0.0140	3.17	0.0209	2.56

* No fitting: $p_{i,t} = \frac{1}{2}\big(R_t(\texttt{rev_5_sector}) + R_t(\texttt{rev_21})\big)_i$.

The weekly IC has a standard deviation of about 0.094 over 2013-2022 (0.109 over 2023-2026), so an IC of 0.016 is tiny compared with its week-to-week noise; only a large number of weeks makes it detectable.

Some of the predictive power may come from effects that every quantitative fund already exploits. To measure what is left, both the predictions and the targets are regressed, week by week, on those effects.

Pure IC

For week t, let F_t be the matrix with one row per stock of $\mathcal{U}_t$ and columns: one indicator per sector, and the ranked `beta_252`, `dollar_vol_21`, `rev_5`, `rev_21`, `mom_12_1` and `vol_63`. Let $e_t^p = p_{\cdot,t} - F_t\widehat{b}_t^p$ and $e_t^Z = Z_{\cdot,t} - F_t\widehat{b}_t^Z$ be the least-squares residuals of the predictions and of the target on F_t. Then

$$\mathrm{IC}_t^{\mathrm{pure}} = \mathrm{corr}\big(R_t(e_t^p),\ R_t(e_t^Z)\big).$$

model	raw $\overline{\mathrm{IC}}$ 2013-22	pure $\overline{\mathrm{IC}}$ 2013-22	t_{NW}	pure 2020-22
reversal rule	0.0190	0.0004	0.16	0.0040
ridge	0.0198	0.0075	2.41	0.0109
LightGBM	0.0161	0.0081	3.23	0.0050

F_t removes the 5- and 21-day reversal, but not the one-day reversal (`rev_1`), the other short-term features, nor the non-linear use that LightGBM can make of the factors in F_t. The table below adds them to F_t, for the LightGBM predictions. About a fifth of the pure IC is one-day reversal. What is left keeps $t > 2.2$ over 2013-2022, but almost all of it comes from 2013-2019: it is about zero over 2020-2022, and about 0.004 with $t \approx 1$ over 2023-2026. The years 2013-2019 are also those with the lowest share of members with a price, so this remainder may be survivorship bias rather than information.

columns of F_t	2013-22	t_{NW}	2013-19	2020-22	2023-26	t_{NW}
sectors and the six factors above	0.0081	3.23	0.0094	0.0050	0.0050	1.14
+ `rev_1`	0.0065	2.65	0.0085	0.0019	0.0044	1.07
+ six short-term features‡	0.0058	2.31	0.0079	0.0008	0.0044	1.03
sectors and the 19 features	0.0058	2.23	0.0086	−0.0008	0.0040	0.94
sectors, the 19 features and their squares	0.0065	2.43	0.0088	0.0011	0.0034	0.81

‡ `rev_1`, `gap`, `intraday`, `rsi2`, `rev_5_sector`, `vol_shock`.

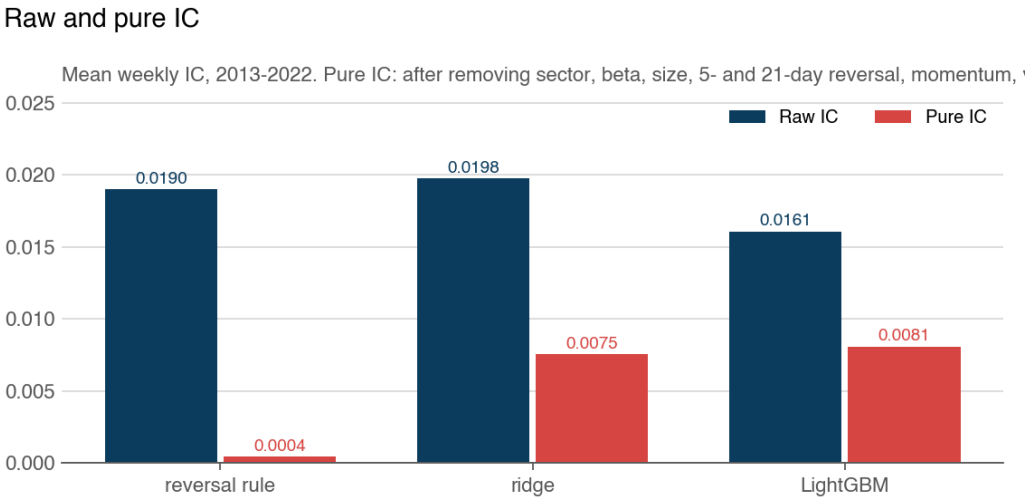


Figure 1: Mean weekly raw and pure IC, 2013-2022.

These variants were computed after the 2023-2026 check; the pure IC with the six factors remains the statistic fixed in advance.

Windows of returns

For a prediction made on $t \in \mathcal{W}$, the IC is measured against sector-relative returns over later windows of sessions, each from the open of session $t + a$ to the open of session $t + b$: week k ($k = 1, \ldots, 4$) is $(a, b) = (5k - 4,\ 5k + 1)$, so week 1 is the target window; day 1 is $(1, 2)$, days 2-5 is $(2, 6)$ and weeks 5-8 is $(21, 41)$. $\mathrm{IC}_{(k)}$ denotes the mean IC over 2013-2022 against week k.

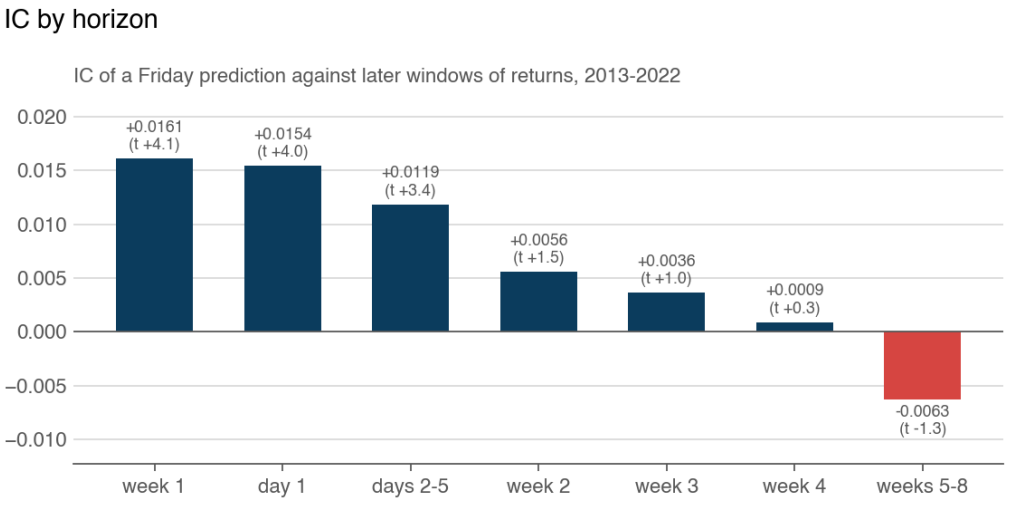


Figure 2: IC of the Friday prediction against later windows of returns, 2013-2022.

Half-life

Week k starts $k-1$ weeks after week 1. The half-life is the time h after week 1 at which the linear interpolation of $k \mapsto \mathrm{IC}_{(k)}$ crosses $\mathrm{IC}_{(1)}/2$:

$$h = k - 2 + \frac{\mathrm{IC}_{(k-1)} - \mathrm{IC}_{(1)}/2}{\mathrm{IC}_{(k-1)} - \mathrm{IC}_{(k)}}$$

for the first $k \geq 2$ with $\mathrm{IC}_{(k)} \leq \mathrm{IC}_{(1)}/2$. Here $h = \frac{0.0161-0.00805}{0.0161-0.0056} = 0.76$ week. Fitting $\mathrm{IC}_{(k)} = \mathrm{IC}_{(1)} e^{-\lambda(k-1)}$ through weeks 1 and 2 gives $\ln 2/\lambda = 0.65$ week.

The information loses half of its strength in three to four sessions. In correlation, the first day alone (0.0154) is almost as informative as the whole week (0.0161). In returns it is not: daily returns vary much less than weekly ones, and the first day holds about 31% of the covariance between the ranked prediction and the week's sector-relative return. A signal this fast has to be traded often, and trading costs become the main issue.

4 Portfolio

To trade less, each stock's prediction is smoothed over the weeks with the following exponential

$$\tilde{p}_{i,t} = \frac{\sum_{k=0}^{K} (1-\lambda)^k \, p_{i,u_k}}{\sum_{k=0}^{K} (1-\lambda)^k}, \qquad \lambda = 1 - 2^{-1/H} \approx 0.29,$$

where $t = u_0 > u_1 > \cdots > u_K$ are the weeks, up to t, in which stock i had a prediction (pandas `ewm` with `halflife=2, ignore_na=True`). The weight of the prediction made k predictions earlier is proportional to $(1-\lambda)^k$, which is halved every H steps since $(1-\lambda)^H = 1/2$. Once the history is long, this is the same as the recursion $\tilde{p}_{i,t} \approx (1-\lambda)\, \tilde{p}_{i,u_1} + \lambda \, p_{i,t}$.

The value $H = 2$ was obtained by rounding a half-life of 1.76 weeks that was measured from the wrong origin (week 1 counted as time 1 instead of 0). It was fixed before the 2023-2026 check and is kept; rounding the correct half-life would give $H = 1$.

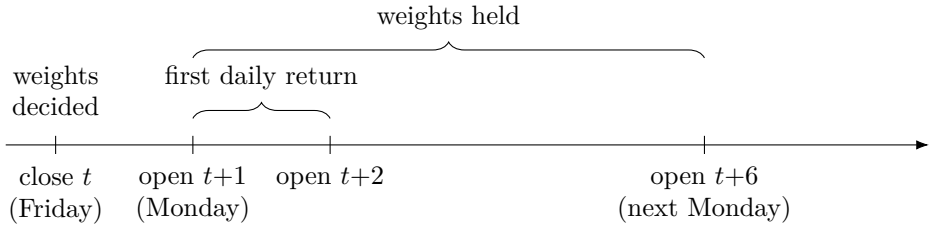
Rank weights

At each week-end session t, for the stocks with a smoothed score that are members with a close at t,

$$w_{i,t} = \frac{\mathrm{rank}_t(\tilde{p}_i) - \bar{r}_t}{\sum_j \left|\mathrm{rank}_t(\tilde{p}_j) - \bar{r}_t\right|}, \qquad \bar{r}_t = \text{mean rank},$$

and $w_{i,t} = 0$ for other stocks, or for every stock if fewer than 300 have a score. Then $\sum_i w_{i,t} = 0$ (dollar-neutral) and $\sum_i |w_{i,t}| = 1$ (half a dollar long and half a dollar short per dollar of capital). The weights are kept unchanged on the following sessions until the next week-end session.

The ranks are taken over all stocks, not within each sector. The book therefore takes sector bets that the model was never trained to predict, since the target is relative to the sector: on average over 2013-2022, $\sum_k \left|\sum_{i:\, s_i(t)=k} w_{i,t}\right| = 0.28$.



Weights decided after the close of t are bought at the open of $t+1$. The open-to-open return of stock i on session d is $g_{i,d} = O_{i,d}/O_{i,d-1} - 1$, and the position held during it is $q_{i,d} = w_{i,d-2}$ (weights of the last week-end session on or before $d-2$).

Let $\widehat{\beta}^s_{i,t}$ be the beta of stock i over the last 252 sessions computed like `beta_252` but with simple returnsinstead of log returns. The portfolio's beta on session t is $\beta^P_t = \sum_i w_{i,t}\widehat{\beta}^s_{i,t}$, and the book sells β^P_t dollars of SPY per dollar of capital (capped at 5), with the same two-session delay as the stocks. Since the betas change every session, the hedge is adjusted every session, not only at the week-end. The hedge makes the book insensitive to the market to the extent that betas are well estimated.

With a cost of c per side (as a fraction of the amount traded) and a borrow fee of 50 basis points a year on short positions, the net daily return of the book on session d is

$$\pi_d = \sum_i q_{i,d}\, g_{i,d} \;-\; c \sum_i |q_{i,d} - q_{i,d-1}| \;-\; \frac{0.005}{252} \sum_i (q_{i,d})_- \;+\; \pi^{\text{hedge}}_d,$$

where π^{hedge}_d is the same expression for the SPY position, with $c = 1$ basis point. Over a period of n sessions:

- the Sharpe ratio is $S = \sqrt{252}\,\bar{\pi}/\hat{\sigma}_\pi$ ($\hat{\sigma}_\pi$: sample standard deviation), and its t-statistic is the Newey-West t of the daily π_d;
- the turnover is $\tau = 252 \cdot \frac{1}{n}\sum_d \sum_i |q_{i,d} - q_{i,d-1}|$, the amount traded per year per dollar of capital;
- the break-even cost is $c^* = \bar{\pi}^{(c=0)} \cdot 252/\tau$, the cost per side at which the mean net return would be zero.

The Sharpe ratio is a ratio of sample moments; reading it as a measure of skill requires the same stationarity assumption as (A1).

The cost term counts only the trades made when the weights change. Keeping the weights $q_{i,d}$ constant from one session to the next is in fact a small rebalancing every session, since prices move in between; those trades are not charged here.

4.1 Results, 2013-2022

	Sharpe ratio for a cost per side of								
construction	0 bp	t	2 bp	5 bp	10 bp	return/yr	vol/yr	τ	c^*
weekly top/bottom 10%†	0.63	2.10	0.40	0.06	−0.50	3.3%	5.2%	59	5.5 bp
smoothed score (main)	0.42	1.39	0.29	0.09	−0.24	1.3%	3.1%	21	6.4 bp

† Long the top 10% of predictions and short the bottom 10%, equal weights, a held stock kept while it stays in the top or bottom 20%.

The Sharpe ratio of the main book is not significantly different from zero ($t = 1.39$). Smoothing cuts turnover by about two thirds for a third of the gross Sharpe ratio. It raises the break-even

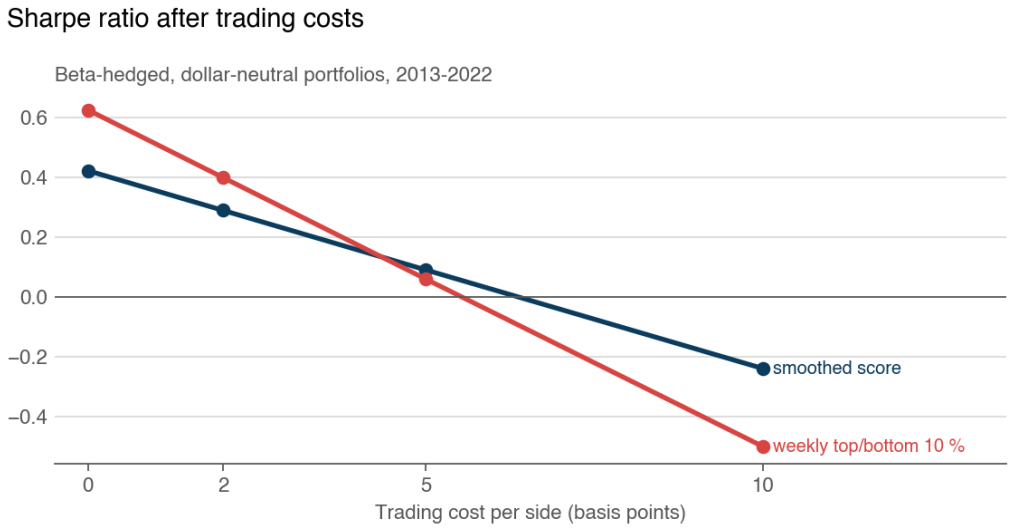


Figure 3: Sharpe ratio of the two books after costs, 2013-2022.

cost, but it does not give a better book at low cost: the two curves cross at about 4.3 basis points, the top/bottom book is better below that, and the smoothed book is better only between 4.3 and 6.4 basis points, where both Sharpe ratios are below 0.15.

5 Check on 2023-2026

The data from January 2023 onwards were not used to build this model: the model, the portfolio construction and the statistic were fixed before this check, which was run once. The period was not untouched, though: earlier tests of other signals in the wider project had already used it twelve times. The bar was therefore set with a Bonferroni correction for 13 tests: $t_{NW} \geq 2.67$ (one-sided level 0.05/13). Two small uses of information from after 2022 remain in the development results: the sector averages of section 3, and the fact that the list of stocks in the data was built in 2026.

2023-2026 ($n = 198$ weeks)	$\overline{\text{IC}}$	t_{NW}	by year (2023, 2024, 2025, 2026)
raw IC	0.0135	1.89	
pure IC (primary statistic)	0.0050	1.14	−0.006, +0.006, +0.014, +0.007

The smoothed book has a Sharpe ratio of 0.26 before costs ($t = 0.53$), 0.15 at 2 basis points and −0.01 at 5. The signal keeps its sign and about 60% of its pure IC, but four years of weekly data cannot separate an IC of 0.005 from zero...

Every weekend, `live/run_weekly.py` predicts the ranking of the current S&P 500 members from Friday's close and appends it to `live/predictions.csv` before Monday's open. The target window ends at the open of the following Monday, so the realised raw and pure IC of a week are appended to `live/results.csv` by the run of the weekend after that, about two weeks after the prediction. A week recorded after Monday's open is flagged and excluded.

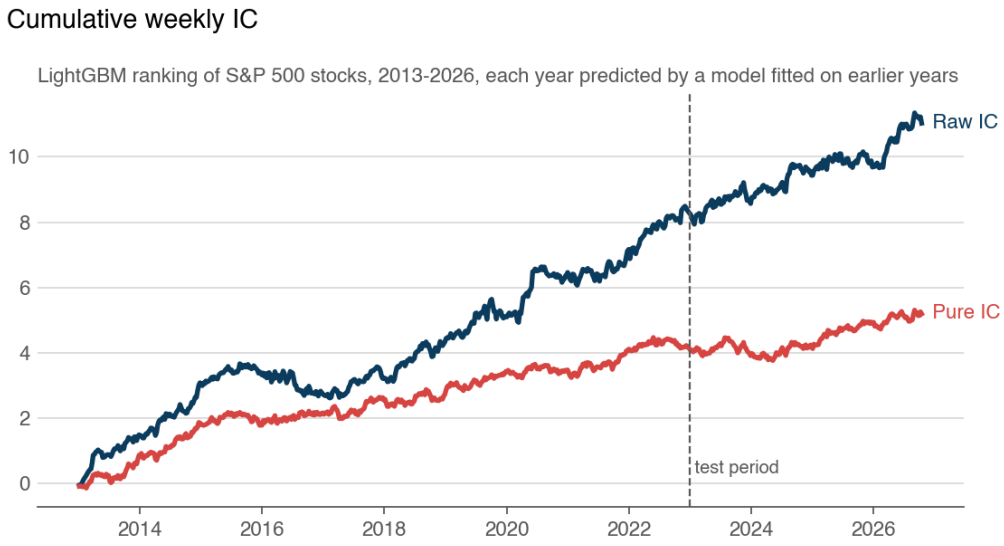


Figure 4: Cumulative sum of the weekly IC since 2013. A steady slope means a positive IC week after week; the dashed line marks the start of the period not used in development.